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Where is the Body in Designing (Through) AI? Frictions and Opportunities in Integrating AI with Soma Design

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Where is the Body in Designing (Through) AI? Frictions and Opportunities in Integrating AI with Soma Design

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Abstract

We make sense of the world around us through our bodies; however, this somatic dimension of meaning-making is often overlooked in the development of AI systems. This workshop (re-)positions the body as central to the design of human-AI interactions by critically exploring the frictions and possibilities that emerge when attempting to incorporate our somatic dimension into the design of predominantly disembodied AI systems. By using self-knowledge as a point of departure to explore the potential of soma-aligned AI as a research territory, our workshop hosts (1) participant-driven discussion on tensions and opportunities between AI design and soma-centric approaches and (2) practical exercises where we together experiment with designing forms of such interactions that are more embodied, sensuous and poetic. We aim to extend these activities beyond the workshop, both by establishing a long-term community of design researchers and through a public *Poetry Jam* event.

CCS Concepts

• **Human-centered computing → Empirical studies in HCI**.

Keywords

HCI, Noticing, LLMs, GenAI, Self-knowledge, Soma design

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1 Motivation and Importance

AI is poised to increasingly mediate our everyday interactions. However, the influence of our *somatic dimension* is often overlooked when devising and designing AI systems that facilitate these mediations [43]. By somatic dimension, we refer to our *corporal knowing*; and particularly our bodily senses and tacit knowledge derived from our interactions with the world. This is not a trivial omission, as our bodies play a pivotal role in our meaning-making process [17, 23]. Bodies are influenced by politics [55], carrying background knowledge about our culture [10], weather [42], stories, affects [54], and so on. Embedded in social situations, bodies shape and are shaped by the contexts in which meaning arises [25], permeating language construction and therefore, shaping how we make sense of the world and ourselves [17, 22]. Our bodies *know their situation*, providing us with tools to act in the world even before we can make a rational sense of our context [45]. This way of knowing is sensuous and generative, which, when attended, opens up to poetic ways of making sense of our world of entanglements and relations [1, 48]. However, there is still a long way to go before this somatic dimension –which is subjective, slow and thrives untamed in uncertainty– is addressed in the design of AI systems.

This workshop invites participants to critically explore the frictions and possibilities that emerge when trying to incorporate our

somatic dimension into the design of AI systems that privilege text-centric, language-oriented approaches where both input and output are seen as disembodied and purely "cognitive" products.

Under the umbrella of *soma design* and its focus on designing from the body [32], and the design of systems with the goal of *promoting self knowledge*, we welcome diverse theoretical and conceptual perspectives interested in interrogating both the tensions and encounters between AI and our somatic dimension. We aim to gather the interest of participants from different subject areas to explore this emerging intersection, including researchers working on AI for (but not limited to) wellbeing, artistic research, embodied interaction, digital humanities, research-through-design, soma design and more-than-human bodies, among others. We also welcome researchers who do not work with AI, but are curious or critical of the developments in this field. Together with participants, we will explore and speculate what soma-centric AI systems might feel and look like, and what qualities they should embody to support respectful, situated, and creative engagements with the self and the world. By engaging in discussions, along with somatic and introspective exercises, we invite participants to situate themselves with their bodies before engaging with AI, and to get inspired by the possibility of applying a poetic and sensorial lens to their own work. Further, we aim to explore the role that our bodies can play in uncovering new human–AI entanglements that foster self-knowledge and reflection.

2 Background

2.1 The somatic dimension of our bodies in AI

Although it is through our bodies that we experience the world in full, in a vast universe of AI papers, our somatic dimension remains mostly underrepresented. We have seen advances in integrating AI and Machine Learning (ML) with different forms of bodily manifestations, particularly movement-based interactions used in artistic applications. A few examples include body mapping visualisation of non-visible features of dance performances [15], image generation through body prompting [44], and body calligraphy [30]. Some have noted how even in highly embodied practices such as singing, the body of voice has been invisibilised in ML and AI practices [16]. Still, the somatic dimension is often disregarded in the design of AI systems, an aspect evident in the limited capacity of algorithmic systems to reciprocate [33] and understand the temporal rhythms, as well as the complexity of our subjectivity [43].

In this workshop, we propose to focus on *self-knowledge* as a starting point and common ground for exploring possible intersections between soma-based approaches and AI. This framing responds to two main reasons: (1) Our somatic dimension is key in how we construct knowledge, and by extension, self-knowledge [22, 54] resonating strongly with soma design and its advocacy for the cultivation of somatic sensibilities as part of the designers' work [32], and (2) it allows for inclusive and expressive interpretations of what self-knowledge entails, which we believe can be potentially relevant to different fields of design research and practice.

General-purpose AI tools, such as ChatGPT and Replika [7, 12], are already being extensively used to gain self-knowledge. This has led, for instance, to research [49] examining how introspective prompts can be utilised to explore existential questions and facilitate personal transformation. In the HCI community, we are seeing an increasing focus on the challenge of ethically and carefully designing with LLMs as materials to support safe, introspective uses. Examples of this research includes the use of LLMs to co-interpret dreams [8], LLM-enhanced journaling tools [29, 34, 41], or chatbots [3] in therapeutic [51], educational [36], or corporate settings to support professional development and self-reflection [2, 35].

Speculative design work in this area [11] that inquires into the space of introspective AI highlights *uncertainty* as a resource that foregrounds the importance of human sense-making in fostering introspection. Uncertainty with AI [4, 6, 26, 31], in particular when foregrounding technical uncertainty, has been described in design research literature, along with the related quality of ambiguity [21]. However, the relationship to bodily knowing in these contributions centres primarily on the *perception* of the designed artefacts, and do not explore the range of bodily expressivity in the interactions proposed. This critique can also be leveraged against work on metaphor in designing with AI, which is likewise biased towards the linguistic and cerebral [40]. On the whole, we therefore argue that there is an overdue consideration of bodily knowing in interaction with AI technologies in the HCI design research community.

2.2 Soma design and self-knowledge

We draw from soma design, a methodology that calls for direct experience with technology materials, promoting a slow and felt approach to engaging and designing interactive systems [32]. The quest for self-knowledge is tightly connected to somaesthetics, a philosophical project that has theoretically influenced soma design, which recognises the body (soma) as a site for sensory appreciation and self-cultivation [50]. In contrast to data-centric views on knowledge generation, soma design prioritises our sensory ways of knowing to make sense of the world [18, 38]. The somatic approach informs both the practice of the designer and the outcome of what is designed, as the cultivation of somatic sensibilities hones our capacity to devise transformative [52] and richer interactive experiences [9, 13, 19, 27, 32, 37], making us better prepared to make ethical decisions in practice [20, 28, 46]. In the context of design for self-knowledge, soma design has inspired several approaches that go beyond the quantification of experience, foregrounding the body, movement, and sensibility as primary focuses [14, 19, 39, 52, 53]. With the disruption of AI in many facets of our everyday lives, we envision a gap where turning to our sensory ways of knowing could inform the design of AI systems for self-knowledge that value ambiguity, slowness and poetics in our somatic meaning-making process [43].

2.3 Opportunities

The fact that LLMs have become popular tools for self-knowledge opens up some possible avenues of exploration, including the design of soma-based systems beyond text, or the recognition of textual language as closely connected to our bodily experience [43]. One of the obstacles for the adoption of AI in soma design is a tendency to situate language as disembodied, as opposed to felt and sensuous [32], even when language *can describe a feeling in a way the scientific method can't* [48]. Embracing this avenue requires expanding the

boundaries of this design program to one that understands language as deeply rooted in our physicality and our interactions with the world [24].

Since designers working on AI for self-knowledge tend to overlook the somatic dimension of our bodies, and soma-designers have yet to position themselves regarding this critical AI disruption, we believe that both communities would benefit from exploring the potential of somatic ways of engaging and designing with AI.

3 Workshop goals

Our workshop aims to carry a body-centric spirit that foregrounds anti-solutionist explorations of self-knowledge. This means supporting qualitative, ongoing, and open-ended approaches that consider bodies and identities as ever-changing and dynamic. Some of our goals for this workshop are:

- To map different tensions and opportunities emerging from the ambition of integrating soma design principles (which value our subjective, somatic dimension) into the design of human-AI interactions.
- To devise new AI/LLMs-facilitated practices that transcend the boundaries of the digital realm, supporting our self-reflective processes in embodied, sensuous, metaphorical, poetic, and fulfilling ways. What would these systems look like/feel like, and how can we ensure they are body-centric?
- To discuss potential ethical challenges emerging when designing AI systems that are soma-centric.

4 Organisers

Our team of organisers comprises a multidisciplinary group of academics with expertise ranging from introspective practices, AI and soma design.

Claudia Núñez-Pacheco is a Senior Lecturer at the Department of Computer Science and Media Technology at Malmö University in Sweden. Her research investigates how to design from self to others, including how bodily ways of knowing are utilised as materials for aesthetic experiences in design. She employs introspective and somatic practices to explore aesthetic and generative qualities in our interactions with technology. http://claudianunezpacheco.com

Pedro Sanches is a Senior Lecturer at the Department of Informatics at Umeå University in Sweden. His research draws on feminist epistemologies to design biodata-driven and AI-based technologies for health and wellbeing, as well as performing arts. He applies speculative and somaesthetic design tactics to consider situated lived experiences when crafting with data and AI as design materials.

Jesse Josua Benjamin is an Assistant Professor at Eindhoven University of Technology. His work combines philosophy of technology and design research and practice, and centres on the underexplored aesthetic potentials of concrete technical attributes of emerging technologies, and the resultant consequences for design research and practice. Having argued for understanding machine learning uncertainty as a design material [5], his current research interrogates the solidifying design conventions around AI technologies.

Iohanna Nicenboim is a postdoctoral researcher at TU Delft, where her work focuses on materialising AI through design. Originally from the Global South, she brings a critical perspective to how AI systems are developed and deployed, emphasising the need for non-extractive, regenerative, and situated approaches. She completed her PhD as a Microsoft Research Fellow, developing a more-than-human design approach to AI that combines posthuman theory with research-through-design practice. She co-edited a special issue of the HCI Journal on the More-than-Human Turn in Design, has chaired two DRS tracks on More-than-Human Design in Practice, and is a chair of the Pictorials track at DIS (2023 and 2026).

Mirjana Prpa is an Assistant Teaching Professor at Northeastern University. Her research interests include leveraging microphenomenology in HCI for understanding the complexity of user experiences [47] and extending it to unfold experiences arising from human-AI persona interactions, spanning from LLM use for persona creation to LLM-based agents in social VR.

Sarah Fdili Alaoui is a Reader at the Creative Computing Institute at the University of the Arts London in interaction design, human-computer Interaction, and dance and technologies. She is a choreographer, a dancer, and a Laban Movement Analyst. Her research investigates the theory, practice, and methods of intersecting technological design with dance-making. Her research methods include artistic research, research-through-design, (auto-)ethnography, phenomenology and action research. She co-founded and organised the MOCO conference.

Michelle Rennerová is an art curator, artist, and PhD student from the University Pompeu-Fabra in Barcelona. In her artistic practice, she explores the intersection of technology and human experience, focusing on social engagement and well-being. Her research focuses on analysing the integration of AI-driven biometric technologies into interactive art installations and their influence on human agency, autonomy, and self-determination.

5 Publication of workshop submissions and availability of workshop outcomes

In the spirit of the design-oriented nature of this workshop –where knowledge generation can take the shape of different artefacts– submissions will not be limited to position papers; we will also accept and encourage participants to submit their expressions of interest (EoI) in various formats, such as pictorials, video submissions demonstrating prototype use or performances, fanzines, and other relevant means.

Workshop materials and submissions will be made publicly available and archived on our event website for future reference. This open documentation will include highlights of workshop discussions, photographs, and expressions of interest from participants who agree to share them. To ensure the accurate representation of our discussions, aspects concerning anonymity and attribution will be shared with participants before making them available on our website.

6 Accessibility

As we gather expressions of interest, we will ask participants to communicate any specific accessibility needs they may have. We will collaborate closely with the workshop chairs to ensure that

everyone has the support they require, with the goal of creating a safe and comfortable workshop experience. Additionally, we will strongly encourage participants to provide alt-text descriptions for the images and figures they might include in their expressions of interest. We will adhere to accessibility design guidelines, ensuring that our website and call for expressions of interest meet the recommended visual standards.

7 Ethics and consent

In terms of ethics, we will exercise a process of "dynamic consent" [56], where participants can feel safe to withdraw or adjust their level of involvement in the activities as they occur throughout the day. As we will engage with somatic exercises to introspect or examine our inner experiences –an later articulate them in text– participants will be informed that they do not need to disclose the context of their texts as well as expressions of interest from participants who agree to share, if participants are comfortable, they can share their personal experiences or concerns related to these activities, which will enrich to our discussions.

8 Workshop activities and length

This workshop comprises two 90-minute sessions, totalling **180 minutes** at the venue, with an additional optional post-event, connecting the outcomes of the workshop with the cultural life in Barcelona. To promote engagement with the somatic exercise and facilitate more focused discussions, we aim to accommodate between 20 and 24 participants.

8.1 Part 0: Before the event

- **Pre-workshop preparations:** The EoI submitted by participants will shape how the workshop is organised. Given the somatic focus of this workshop, we encourage the submission of various formats of expressions of interest to participate, such as (but not limited to) position papers (2-4 pages), pictorials, video performances, prototypes, etc. These submissions should frame their contributions broadly as body-centric, such as (but not limited to),
  - Theoretical approaches bridging AI and the body/soma design,
  - Creative uses and practices that use AI and movement/ embodiment.
  - We are also interested in approaches that explicitly contest the need for AI in favour of other body-centric practices.

  We will share the submissions with the participants, which will be hosted on our website two weeks before the workshop, inviting them to familiarise themselves with the materials. We will propose thematic groups that will work together during the event, based on shared interests, assessed from the EoIs. This will be done to foster longer-term collaborations among participants. We will additionally create a Slack channel and host an online, pre-workshop meeting where we will invite participants to introduce themselves and their research interests, this way, saving time to engage directly in the hands-on activities.

8.2 Part 1: Introduction and somatic exercise with LLMs

- **Opening [10 mins]:** Introduction to the objectives and schedule of the workshop. We will also discuss our policy of dynamic consent, which allows participants the freedom to withdraw from the exercises as they unfold, even if they have previously approved joining.
- **Working on participants' submissions: [20 mins]** In small groups, participants will be encouraged to discuss their reasons to join the workshop, their expectations and expressions of interest. We will invite them to interrogate how the body is present or absent in their submissions.
- **Somatic exercise: Guided Focusing and felt prompting [60 mins]** The organisers will facilitate activities [43] to attune participants to their interoceptive awareness. These will serve as starting points for participants to imagine their own versions of poetic and soma-based AI systems for self-knowledge, as well as encouraging reflection on ethical considerations for the design of body-centric AI systems.
  - **Focusing:** We will follow the protocol we have introduced here [43] and depicted in Figure 1 below. It begins with a guided exercise that introduces the question *What is it like to be me today?*, which poses self-knowledge as a dynamic process. Participants then take notes with the emerging sensations during the session.
  - **Feeling the prompt as a poetry dialogue:** As a next step, participants will be encouraged to engage poetically with LLMs and potentially other AI-based systems to engage with their introspective notes, potentially by creating haikus or similar forms of short-form poetry. To ensure ownership of this process for both organisers and participants, as well as to address privacy concerns, we will provide a custom web-based LLM interaction to participants for the duration of the workshop. This will feature an API connection to a LocalAI[1] model instance hosted at the Eindhoven University of Technology's DataFoundry provision[2] to guarantee control over data processing. There will be no storing of participant interactions or any server-side storing of cache or cookie files.
- **Break** To rest, refresh and socialise.

8.3 Part 2: Discussions

- **Sharing and devising new opportunities [30 min]:** Within the same groups, participants who are willing to share will read their poems, which will inform our discussions of findings in this exercise. Then, they will map what they have discovered from the exercise, including how this poetic approach to self-knowledge and AI can inspire new ways of designing poetic, multimodal interactions, based on the participants' interests. Based on our discussions and poetic activity, we will speculate and ideate together on what soma-based AI systems that take into account our bodily knowledge look and feel like. We will discuss how these requirements might bring frictions with existing ways of designing AI systems.

[1] https://github.com/mudler/LocalAI, accessed 22/09/2025.
[2] https://data.id.tue.nl/, accessed 22/09/2025.

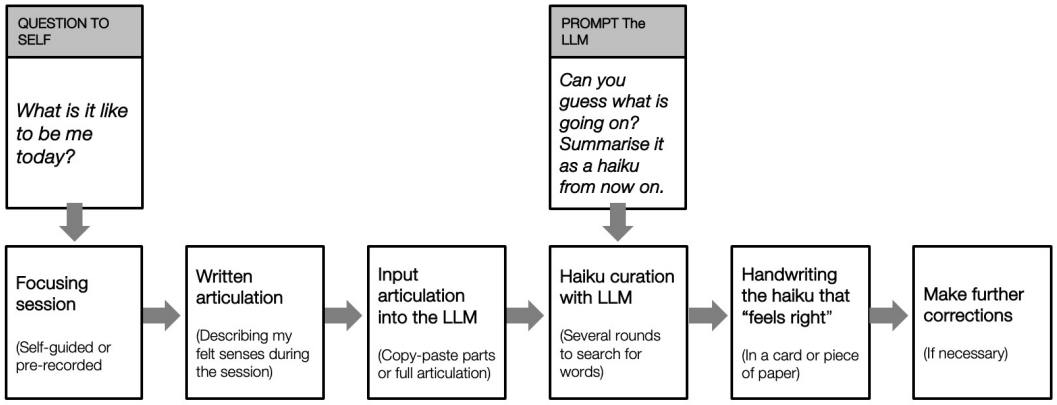


Figure 1: The study protocol

- **Categorising potential frictions and encounters between a possible integration of AI and soma design: [30 minutes]** We will discuss and articulate our impressions in mind maps completed collaboratively and taking turns. We will cover this in three rounds, in a world cafe setting, by discussing the following questions: (1) What are the challenges, and what can be learn from the seemingly incompatible nature of AI and soma design? (2) What is the current state of the body in the design of AI for self-knowledge, and how could soma design contribute (or not)? (3) Which theories could help us inform our integration attempts? These questions will be posed in different sections of the room, either on the walls or tables. We will provide large pieces of paper and Post-it notes for each section, allowing participants to leave traces of their discussions as they move.
- **Collective debriefing and closing [30 minutes]:** We will articulate our findings together and close the session, inviting participants who are interested to join an optional Poetry Jam session to take place after hours.

8.4 Voluntary post-workshop event: Poetry Jam

We aim to extend this event beyond the conference boundaries, by taking part in the cultural life in Barcelona. After the official session is finished, interested participants will have the chance to join us in a non-competitive poetry jam session, where we will collectively read aloud and transform some of the poetry crafted during the workshop. We will also extend this invitation to the rest of the CHI community. In doing so, we hope to foster the development of a community of researchers and practitioners interested in soma design and AI, beyond the workshop and in an informal setting. Beyond reading, we will support the improvisational communication of these poems through music, visual art, performance, and other forms. Michelle Rennerová, curator, artist, and co-organiser, is securing a local creative venue for us to continue discussing, sharing, and working together to materialise the workshop outcomes in a collaborative, artistic way, as afforded by poetry.

9 Post-workshop plan and outcomes

The poetic texts collected during the workshop will be compiled into a zine to be submitted at ACM DIS 2026, where all participants contributing to them will be listed as authors. Along with the participants, we will explore further avenues to disseminate this material through digital and physical formats, including the possibility of collaborating in future publications derived from our discussions.

10 Call for Participation (250 words)

We make sense of the world around us through our bodies; however, this somatic dimension of meaning-making is often overlooked in the development of AI systems. Through poetic engagements with AI and our bodily ways of connecting with self-knowledge, this long workshop invites participants to critically explore the frictions and opportunities that emerge when bridging seemingly contrasting practices, such as soma design and the design of AI systems. We welcome expressions of interest (EoI) that explore this topic from a variety of perspectives, such as (but not limited to):(1) Theoretical approaches bridging AI and the body, (2) creative practices that employ AI and embodiment, (3) positions that contest the need for AI in favour of other body-centric practices, etc. EoIs can take various formats, including position papers, pictorials (2-4 pages), video prototypes and so on. Submissions should be sent to claudia.nunezpacheco@mau.se, and those selected will be made available on our website http://claudianunezpacheco.com/soma-ai-CHI2026 upon the participant's agreement. We will select participants who align with the body-centric AI aims and whose submissions have the potential to contribute to the discussions. At least one author of each accepted submission must attend the conference. We aim to gather around 20 participants from different areas, including (but not limited to) AI for wellbeing, artistic research, embodied interaction, digital humanities, RtD, soma design, and more-than-human bodies, among others.

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